

Hugo FRANCOIS

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Ferney-Voltaire 01210 · Junior Cybersecurity Engineer · Applied Cryptography, PKI & Post-Quantum Security

EDUCATION

2025 - 2026	CENTRALESUPÉLEC Double Degree in Engineering, Cybersecurity specialization	RENNES, FRANCE
2025	STANFORD UNIVERSITY Cryptography I — Certificate	ONLINE
2024	UNIVERSITY OF CALIFORNIA, LOS ANGELES (UCLA) Study Abroad Semester: Cybersecurity Fundamentals, TCP/IP & Management	LOS ANGELES, USA
2021 - 2026	ESME SUDRIA Engineering Degree (Diplôme d'Ingénieur), Cybersecurity major	PARIS, FRANCE

EXPERIENCE

May 2026 - Present	ALMOND Cyber Technology & Transformation - infrastructure security and post-quantum transition <i>Cybersecurity Consultant Intern</i> - Leading technical work on Almond's PKI migration toward a dual classical/post-quantum architecture, covering trust chains, identities, certificate lifecycle and deployment readiness. - Designed and deployed Azure PoCs for dual PKI, an ML-DSA-65 CA, hybrid mTLS/PQC authentication, and TLS 1.3 hybrid key exchange using X25519MLKEM768. - Engineered an Nginx/FastAPI gateway that validates classical and PQC credentials before authorising access to legacy backends. - Built Docker, Bash and Python test automation for certificate issuance/revocation, CRLs and TLS interoperability; produced PKI migration and crypto-agility recommendations.	SÈVRES, FRANCE
Jan 2026 - Present	LORNE & VALE Legal-tech company automating client intake for law firms in France and the UK <i>Co-Founder & Managing Director</i> - Co-founded the company and co-lead workflow automation, client onboarding and product delivery. - Own security and infrastructure design, including restricted-access architecture, data-minimisation controls and post-quantum protection features for sensitive case data.	PARIS, FRANCE
Jun 2025 - Sep 2025	EDF Nuclear Protection Division - cybersecurity governance and operational monitoring <i>Cybersecurity Intern</i> - Designed a SharePoint + Power BI cybersecurity steering solution to track internal compliance, KPIs and supporting evidence in real time. - Automated follow-up and evidence centralisation, replacing Excel-based tracking and delivering an estimated annual operational value above EUR 50K.	FRANCE
2022 - 2023	SEPTODONT Pharmaceutical R&D environment <i>Engineering Intern / R&D Research Assistant</i> Supported laboratory R&D and experimental work; engineering internship extended into a fixed-term research role.	SAINT-MAUR-DES-FOSSÉS, FRANCE

SELECTED TECHNICAL PROJECTS

PQC-OnionPKI - Hybrid PKI & Post-Quantum Email Signing | *Raspberry Pi · Tor · Postfix · OpenSSL-OQS · ML-DSA-65 · Python*
Designed a hardened hybrid PKI exposed through a Tor onion service, combining classical trust for ecosystem compatibility with server-side ML-DSA-65 signatures on outgoing email.
Integrated deterministic signing/verification without client modification; measured key generation, signing and verification at about 20 ms on Raspberry Pi 4.

Final-Year Engineering Project - Experimental Symmetric Cryptography | *Cryptography · Algorithm Design*
Designed an experimental symmetric encryption algorithm based on bigram-frequency analysis across multidimensional grids, extending an earlier personal research project.

More projects : hugofrancois.fr

SKILLS & ADDITIONAL INFORMATION

Skills: PKI/X.509 · TLS 1.3 · RSA/ECC · ML-KEM · ML-DSA · hybrid PQC · Azure · Linux · Docker · OpenSSL/liboqs · Python · Bash · Power BI · Windows · Linux · IOS

Languages: French (native) · English (C1, IELTS 7.0/9) · German (B1) · Italian (A2)

Certificates: Stanford University - Cryptography I (2025) · TryHackMe - Complete Beginner, Jr Penetration Tester, Web Fundamentals (2024)

Leadership & Sport: Vice-President, ESME High-Level Athletes Association (2023-2025) · UCLA Official Badminton Team, California university champion (2024) ·